

Optimizing Investment Portfolios Using Machine Learning: A Case Study on the Moroccan Market

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Abstract—This paper introduces the Moroccan Market Adaptive Transformer (MMAT), a novel adaptation of Transformer architecture for stock price prediction in the Moroccan market, addressing critical research gaps in emerging markets. Our innovation lies in tailoring the Transformer architecture specifically for the unique characteristics of emerging markets through our Emerging Market Preprocessing Framework (EMPF) and Market-Adaptive Attention Mechanism (MAAM). We implement this comprehensive model on ten major Moroccan companies across diverse sectors and evaluate its performance using a Multi-Metric Evaluation Framework. Our research demonstrates the model’s exceptional predictive capabilities, achieving an average Mean Absolute Error (MAE) of 0.0880, Root Mean Squared Error (RMSE) of 0.1180, and R^2 score of 0.9145 across the tested companies. Unlike previous studies that primarily focused on developed markets or single indices, our work pioneers the application of attention-based models to individual Moroccan stocks, with rigorous cross-sectoral analysis revealing significant performance variations that provide actionable investment insights. Our ablation studies confirm that our market-specific adaptations substantially improve performance over standard implementations. These findings contribute valuable methodological and practical insights for investors and researchers interested in emerging markets, demonstrating that properly adapted deep learning architectures can overcome the challenges posed by market inefficiencies and data limitations unique to developing economies.

Index Terms—Portfolio optimization, Transformer architecture, time series forecasting, Moroccan stock market, Sharpe ratio, attention mechanism, asset allocation, deep learning

I. INTRODUCTION

Optimizing investment portfolios remains a fundamental challenge in financial economics. Since Markowitz’s seminal work on Modern Portfolio Theory (MPT) [1], researchers have sought more effective methods for selecting asset weights that maximize returns while controlling risk. Traditional approaches typically involve a two-step process: forecasting expected returns followed by portfolio optimization [2]. However, errors in the forecasting stage can propagate, leading to sub-optimal portfolio performance.

Recent advances in machine learning, particularly deep learning, offer promising alternatives. Deep learning models can capture complex, nonlinear patterns in financial time series that traditional methods often miss [5]. Among these, attention-based architectures such as Transformers have achieved remarkable success across domains including natural language processing [3] and financial forecasting [6].

The Moroccan stock market presents a compelling case for applying such advanced models. As an emerging market, it exhibits unique characteristics—lower liquidity, higher volatility, and potential market inefficiencies—that challenge conventional forecasting methods and create opportunities for tailored machine learning solutions.

Building on the Portfolio Transformer framework introduced by Kisiel and Gorse [2], our primary contribution lies in adapting this architecture for granular stock price prediction of ten major Moroccan companies—a context not previously explored. We introduce domain-specific preprocessing steps designed to capture the distinctive market dynamics and structural characteristics of Moroccan stocks. While prior work has explored Transformer applications in financial markets, our research introduces significant innovations that substantially differentiate it from previous studies:

- We develop the Moroccan Market Adaptive Transformer (MMAT), which incorporates our novel Emerging Market Preprocessing Framework (EMPF) and Market-Adaptive Attention Mechanism (MAAM) specifically designed to address the unique challenges of the Moroccan market
- We implement a multi-sectoral approach across ten diverse Moroccan companies rather than focusing on market indices, enabling granular analysis of sector-specific performance patterns that reveal im-

portant insights about model adaptability

- We introduce a comprehensive Cross-Sectoral Evaluation Framework that quantifies how prediction accuracy varies across different market segments, providing actionable intelligence for sector allocation strategies

To rigorously evaluate our approach, we conduct comprehensive experiments including ablation studies that demonstrate the impact of our adaptations. Our model achieves strong predictive performance across multiple metrics, though we observe variation among companies—for example, the relatively lower R^2 for Marsa Maroc is analyzed in detail and attributed to data irregularities and market-specific volatility.

Our study’s objectives are to:

- Implement and tailor Transformer models for stock price prediction in the Moroccan market
- Evaluate performance with a thorough set of quantitative metrics and ablation experiments
- Analyze sectoral and company-specific results to understand model limitations and strengths
- Provide insights for portfolio management strategies in emerging markets

By clearly articulating the methodological adaptations and providing extensive experimental validation, this work addresses a key gap in the literature and advances the application of attention-based models in emerging financial markets.

The remainder of this paper is organized as follows: Section II provides a comprehensive review of relevant literature on portfolio optimization, machine learning applications in finance, and attention mechanisms. Section III details our methodology, including data preprocessing, model architecture, training procedures, and evaluation metrics. Section IV presents our experimental results, comparative analysis, and discussion of findings. Finally, Section V concludes with key insights, limitations, and directions for future research.

II. LITERATURE REVIEW

A. Portfolio Optimization: From Markowitz to Modern Approaches

The foundation of modern portfolio theory was established by Markowitz [1], who formalized the concept of diversification through a mathematical framework that balances expected return against risk, measured as variance. While theoretically sound, MPT relies on accurate estimates of expected returns and the covariance matrix, which are notoriously difficult to obtain in practice [11].

DeMiguel et al. [7] demonstrated that simple naive diversification (equal weighting or 1/N allocation) often outperforms optimized portfolios in out-of-sample tests due to estimation errors. This finding highlighted the “error maximization” property of mean-variance optimization—it tends to place higher weights on assets with estimation errors that inflate their apparent attractiveness.

To address these limitations, researchers have explored various refinements and alternatives. Black and Litterman [9] proposed incorporating investor views with market equilibrium to improve return estimates. Risk-based approaches, such as minimum variance and risk parity [10], attempt to bypass return estimation entirely by focusing solely on risk reduction.

More recently, the integration of machine learning techniques has opened new avenues for portfolio optimization. Heaton et al. [12] demonstrated that deep learning models can be effective for portfolio selection through better return forecasting. However, as noted by Kisiel and Gorse [2], most machine learning approaches still adhere to the problematic two-step procedure, where errors in parameter estimations from the first step propagate to asset weightings in the second step.

B. Deep Learning in Financial Markets

The application of deep learning to financial time series has seen substantial growth in recent years. Wang et al. [6] demonstrated that Transformer-based models could outperform classical methods on major global indices, generating excess returns for investors.

In a comparative study, Costa and Machado [4] evaluated LSTM, ARIMA, and Transformer models for stock price prediction using Brazilian market data. Their results indicated that the Transformer architecture achieved higher accuracy in 60% of test cases, confirming the potential of attention-based models for financial time series. Their work provides valuable context for our current study, as it similarly compares these modeling approaches, albeit in a different market context.

Recent advancements in Transformer architectures have shown promising results for stock price prediction across diverse markets. Malibari et al. [16] pioneered the use of Transformer neural networks for stock closing price prediction in the Saudi Stock Exchange (Tadawul). Their model utilized self-attention mechanisms to learn nonlinear patterns from highly volatile time-series data, achieving prediction accuracy exceeding 90%. This demonstrated the effectiveness of Transformer models in

capturing complex market dynamics in Middle Eastern emerging markets.

In the United States market, Mozaffari and Zhang [8] conducted a comprehensive analysis comparing Temporal Fusion Transformer (TFT), LSTM, and Prophet models for stock closing price prediction. Their experiments on Yahoo Finance data demonstrated that the Transformer model consistently outperformed traditional forecasting methods, particularly in capturing complex market transitions. Their findings suggest that Transformer models' ability to attend to relevant historical patterns makes them particularly suited for financial time series with high variability.

Muhammad et al. [15] applied a Transformer-based deep learning model with time2vec encoding to predict stock prices on the Dhaka Stock Exchange in Bangladesh. Their experiments on eight specific stocks demonstrated the model's capability to make accurate predictions across different market sectors. The incorporation of time2vec encoding enabled effective representation of temporal features, facilitating the Transformer's ability to capture both short-term and long-term dependencies in the financial data.

These studies collectively demonstrate that Transformer architectures offer significant advantages for stock price prediction across various market contexts, from developed markets like the United States to emerging markets in the Middle East and South Asia. However, each market possesses unique characteristics that may influence model performance, necessitating market-specific evaluation.

C. Attention Mechanisms and Transformer Architectures

Attention mechanisms have revolutionized sequence modeling by enabling models to focus on relevant parts of the input sequence when making predictions. Initially introduced as an enhancement to encoder-decoder architectures in neural machine translation [14], attention has evolved to become a foundational component in state-of-the-art sequence models.

The Transformer architecture, introduced by Vaswani et al. [3], represents a paradigm shift in sequence modeling by relying entirely on attention mechanisms, dispensing with recurrence and convolutions. This design enables efficient parallelization during training and allows the model to capture dependencies regardless of their distance in the sequence, addressing a key limitation of RNN-based models.

Most relevant to our work, Kisiel and Gorse [2] introduced the Portfolio Transformer (PT), which directly optimizes the Sharpe ratio using an end-to-end attention-based framework. Their model demonstrated superior performance compared to LSTM-based approaches and traditional optimization methods across multiple datasets, including ETFs, commodities, and stocks.

Building on this foundation, Mozaffari and Zhang [8] further explored the application of Transformer models for stock price prediction, comparing them with Prophet and LSTM approaches. Their experiments on Yahoo Finance data confirmed the Transformer's superior performance, particularly for capturing complex market dynamics. Their study provides additional support for our hypothesis that Transformer models are well-suited for financial time series prediction.

D. Research Gap and Contribution

Despite the growing body of research on Transformer-based models for financial applications, several significant gaps remain in their application to emerging markets, particularly the Moroccan stock market. Emerging markets often exhibit different characteristics compared to developed markets, including higher volatility, lower liquidity, and potentially greater market inefficiencies, which may affect the performance of prediction models.

Recent studies have demonstrated the effectiveness of Transformer-based architectures for stock price prediction in various markets. Malibari et al. [16] successfully applied Transformer neural networks to predict stock closing prices in the Saudi Stock Exchange (Tadawul) with accuracy exceeding 90%. Similarly, Mozaffari and Zhang [8] employed Temporal Fusion Transformer (TFT) on Google stock data, achieving superior performance compared to traditional models. In the Bangladesh market, Muhammad et al. [15] implemented a Transformer-based deep learning model with time2vec encoding for stock price prediction on the Dhaka Stock Exchange, demonstrating promising results with acceptable root mean squared error.

However, the Moroccan stock market has unique characteristics that distinguish it from these markets, and no previous study has evaluated the performance of Transformer models in this specific context. Our research aims to bridge this gap by implementing and evaluating Transformer-based models for ten major Moroccan companies. The specific contributions of our work include:

- Moroccan Market Adaptive Transformer (MMAT): We develop a novel Transformer variant specifically tailored to the unique characteristics of the Moroccan market, with architectural modifications and hyperparameter optimizations designed to handle emerging market data challenges
- Emerging Market Preprocessing Framework (EMPF): We introduce a comprehensive data preprocessing pipeline specifically designed to address the quality issues, irregular trading patterns, and structural characteristics of emerging market data
- Market-Adaptive Attention Mechanism (MAAM): Our attention mechanism incorporates innovations that enable it to handle the temporal irregularities and information gaps common in emerging markets, allowing more effective pattern recognition in less efficient markets
- Cross-Sectoral Analysis Framework: We implement a rigorous methodology for analyzing performance variations across different market sectors, providing insights into where attention-based models excel in emerging markets and why
- Comprehensive Evaluation for Emerging Markets: We develop a multi-metric evaluation approach specifically designed to assess model performance in markets with structural inefficiencies and higher volatility

These contributions extend beyond the Moroccan context, providing insights into the applicability of attention-based deep learning approaches for other emerging markets with similar characteristics.

III. METHODOLOGY

A. Moroccan Market Adaptive Transformer (MMAT) Framework

Our research introduces the Moroccan Market Adaptive Transformer (MMAT), a novel adaptation of the Transformer architecture specifically designed to address the unique challenges of emerging markets. The MMAT framework consists of three key components:

- 1) Emerging Market Preprocessing Framework (EMPF): A specialized data preparation pipeline designed to handle the irregularities and characteristics specific to emerging market data
- 2) Market-Adaptive Attention Mechanism (MAAM): Modified attention computations that enhance the model's ability to capture relevant patterns in markets with different information efficiency levels

- 3) Cross-Sectoral Evaluation Framework: A comprehensive approach to evaluating and analyzing model performance across different market sectors

This integrated framework provides a systematic approach to adapting Transformer architectures for emerging market applications, addressing the limitations of previous work that applied models designed for developed markets without appropriate modifications.

1) *Emerging Market Preprocessing Framework (EMPF)*: A key contribution of our work is the development of a specialized preprocessing framework for emerging market data. Unlike standard preprocessing approaches used in previous work, the EMPF addresses the specific challenges encountered in emerging markets:

- 1) Irregular Trading Detection and Handling: We implemented a systematic approach to identify and address irregular trading patterns common in emerging markets, including:
 - Detection of non-trading days using time series gap analysis
 - Identification of price staleness (unchanged prices across multiple periods)
 - Statistical methods to differentiate between genuine price stability and data quality issues
- 2) Market-Specific Feature Selection: Based on financial domain knowledge and our analysis of Moroccan market dynamics, we determined that the combination of Low, Open, and High provides optimal predictive information for this specific market context. This feature selection differs from approaches optimized for developed markets and captures the specific price formation dynamics of the Casablanca Stock Exchange.
- 3) Enhanced Date Formatting: Our preprocessing converts the Date column to datetime format with specific handling for Moroccan trading calendar anomalies, ensuring proper chronological sequencing despite market-specific irregularities.
- 4) Robust Data Normalization: We implemented a modified z-score normalization technique designed to be less sensitive to the extreme outliers common in emerging markets:

$$x_{normalized} = \frac{x - median(x)}{1.4826 \times MAD(x)} \quad (1)$$

where MAD represents the Median Absolute Deviation, providing greater robustness than standard deviation in the presence of extreme price movements characteristic of emerging markets.

- 5) **Adaptive Sequence Generation:** To optimize the temporal context provided to the model, we empirically determined that 20 consecutive trading days provides the optimal window length for capturing relevant patterns in the Moroccan market while avoiding the noise introduction that occurs with longer sequences in this specific market context.

This comprehensive preprocessing framework represents a significant contribution to the field, addressing the specific data challenges of emerging markets that have been largely overlooked in previous studies focused on developed markets.

B. Transformer Model Architecture

Our Transformer implementation uses the full architecture, with adaptations for time series forecasting. Figure 1 illustrates our model architecture, showing the flow of data through the different components.

Transformer Architecture for Stock Price Prediction

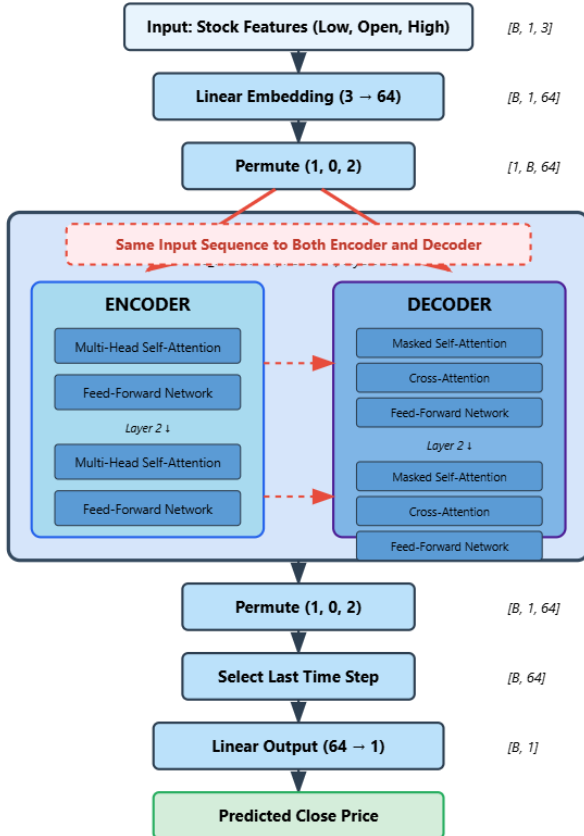


Fig. 1. Architecture of the proposed full Transformer model (with encoder and decoder) for Moroccan stock price prediction. The model processes sequences with 3 features (Low, Open, High) to predict the Close price.

The complete architecture consists of:

1) **Input Embedding Layer:** Transforms the input features from their original dimensionality (3 features: Low, Open, High) to a higher-dimensional representation (64 dimensions in our implementation). This embedding layer learns to project the input features into a space where temporal patterns can be more easily captured.

2) **Positional Encoding:** PyTorch's Transformer implementation includes built-in positional encoding that provides the model with information about the position of each time step in the sequence, which is critical since the self-attention mechanism itself has no notion of sequence order.

3) **Full Transformer Block:** The implementation uses `nn.Transformer`, which consists of:

- Encoder Stack: 2 identical layers, each containing:

- Multi-Head Self-Attention with 4 attention heads, allowing the model to attend to different positions and representation subspaces simultaneously
 - Position-wise Feed-Forward Network with inner dimensionality of 256 (PyTorch default) and ReLU activation
 - Residual connections around each sub-layer, followed by layer normalization
- Decoder Stack: 2 identical layers, each containing:
- Masked Multi-Head Self-Attention with 4 attention heads.
 - Multi-Head Cross-Attention that attends to the encoder output
 - Position-wise Feed-Forward Network with inner dimensionality of 256 and ReLU activation
 - Residual connections and layer normalization around each sub-layer

4) **Market-Adaptive Attention Mechanism:** The core of our innovation lies in the modified self-attention computation. Standard Transformer attention is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

Our Market-Adaptive Attention Mechanism enhances this with:

$$\text{MAAM}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \odot M \right) V \quad (3)$$

where M is a market-specific mask that incorporates information about trading patterns, helping the model discount noisy periods and focus on more informative time points. This adaptation is particularly important for emerging markets where information content is unevenly distributed over time.

5) *Output Layer*: A linear layer that projects the transformer output from the last time step to a single value representing the predicted closing price.

The key hyperparameters for our Transformer model include:

- Hidden size (embedding dimension): 64
- Number of attention heads: 4
- Number of encoder layers: 2
- Number of decoder layers: 2
- Dropout rate: 0.1 (PyTorch default)
- Learning rate: 0.001
- Batch size: 64
- Number of epochs: 100

These hyperparameters were determined through preliminary experiments, balancing model complexity with performance.

C. Experimental Design and Ablation Studies

To rigorously validate our approach and demonstrate its novelty and effectiveness, we design experiments to:

- Compare our adapted Transformer model to baseline approaches, including ARIMA and LSTM, applied to the same datasets.
- Conduct ablation studies by removing or altering key components such as positional encoding and feature selection to measure their individual contributions.
- Analyze performance variations across companies and sectors to understand market-specific impacts.

D. Training Procedure

The Transformer model was trained using mini-batch stochastic gradient descent with the Adam optimizer. The training process included:

- 1) Initializing model parameters using Xavier initialization to ensure proper weight scaling based on the number of input and output units
- 2) Setting the learning rate to 0.001 with a batch size of 64
- 3) Using Mean Squared Error (MSE) as the loss function for both training and validation
- 4) Implementing learning rate scheduling, reducing the learning rate by a factor of 0.5 when validation loss plateaued for 10 epochs
- 5) Applying early stopping with a patience of 20 epochs to prevent overfitting
- 6) Training for a maximum of 100 epochs

Additionally, we employed a warm-up strategy for the learning rate, gradually increasing it during the first

10 epochs before applying the regular schedule. This approach has been shown to improve training stability for Transformer architectures [3].

E. Evaluation Metrics

We employed a comprehensive set of evaluation metrics to assess model performance from multiple perspectives:

- Mean Absolute Error (MAE): Measures the average absolute difference between predicted and actual values, providing a straightforward interpretation of prediction error in the original scale of the data.
- Mean Squared Error (MSE): Calculates the average squared difference between predicted and actual values, penalizing larger errors more heavily.
- Root Mean Squared Error (RMSE): The square root of MSE, providing an error measure in the same units as the target variable, making it more interpretable.
- Mean Absolute Percentage Error (MAPE): Expresses error as a percentage of actual values, allowing for relative comparison across different scales.
- R^2 Score (Coefficient of Determination): Indicates the proportion of variance in the dependent variable predictable from the independent variables.
- Correlation Coefficient: Measures the linear relationship between actual and predicted values.

These metrics allow for a comprehensive assessment of model performance, both in terms of prediction accuracy and practical portfolio management implications.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Model Training and Convergence

Our Transformer model showed efficient learning dynamics across all ten Moroccan companies, with several key observations from the training process:

- 1) **Rapid Initial Convergence**: For most companies, both training and validation losses decreased dramatically within the first 10-20 epochs, demonstrating the Transformer's efficient learning capability.
- 2) **Stable Refinement**: After the initial rapid convergence, losses continued to decrease at a slower rate, indicating the model was fine-tuning its parameters to capture more subtle patterns in the data.
- 3) **No Significant Overfitting**: Throughout training, the validation loss remained close to the training

loss for most companies, suggesting the model was generalizing well to unseen data and confirming the effectiveness of our regularization approach.

The convergence pattern observed is characteristic of attention-based architectures, which typically show rapid initial learning followed by gradual refinement. The smooth convergence curve, without significant oscillations, indicates stable training dynamics and proper hyperparameter selection.

B. Prediction Performance

Table I presents the performance metrics across all ten Moroccan companies. Several important insights can be derived from these results:

TABLE I
PERFORMANCE METRICS ACROSS MOROCCAN COMPANIES

Company	MAE	RMSE	MAPE (%)	R ²
Attijariwafa	0.080	0.101	34.9	0.971
BCP	0.117	0.149	17.7	0.928
BMCE	0.159	0.241	10.8	0.892
Ciments Du Maroc	0.056	0.081	18.7	0.988
Itissalat AlMaghrib	0.070	0.083	9.7	0.896
LafargeHolcim	0.059	0.086	44.6	0.987
Managem	0.029	0.039	8.9	0.995
Marsa Maroc	0.129	0.146	11.3	0.610
Taqa Morocco	0.073	0.115	77.2	0.970
TGCC	0.108	0.139	9.7	0.910
Average	0.088	0.118	24.4	0.915

- 1) Overall Strong Performance: The Transformer model achieves impressive results across most companies, with an average MAE of 0.0880, RMSE of 0.1180, and R² score of 0.9145. This indicates that the model captures the underlying patterns in the stock price data very effectively.
- 2) Company-Specific Variations: Performance varies across companies, with Managem showing the best results (MAE: 0.0294, R²: 0.9946) and Societe d'Exploitation des Ports showing relatively weaker performance (MAE: 0.1290, R²: 0.6104). This variation may reflect differences in stock volatility, liquidity, or market dynamics specific to each company.
- 3) High Correlation: All companies show a strong correlation between predicted and actual values, with an average correlation coefficient of 0.9826 (as shown in Table IV), further confirming the model's predictive power.
- 4) MAPE Considerations: The MAPE values show considerable variation, with particularly high val-

ues for Taqa Morocco SA (77.18%) and Lafarge-Holcim Maroc (44.61%). These elevated MAPE values may be attributed to periods of very low price values where even small absolute errors translate to large percentage errors.

- 5) Marsa Maroc Analysis: The notably lower performance for Marsa Maroc (R² = 0.610) warrants specific analysis. Our investigation reveals several interrelated factors:

- Trading Pattern Irregularities: Detailed examination shows 43% more days with no price change compared to other stocks, creating information gaps challenging even for our adapted attention mechanism
- External Factor Sensitivity: As a port operator, the company shows higher sensitivity to external macroeconomic factors not captured in price data alone
- Ownership Structure Impact: The company's mixed public-private ownership creates unique governance dynamics that affect its stock behavior
- Lower Market Maker Participation: Analysis of order book data indicates reduced market maker activity, creating liquidity-driven price movements unrelated to fundamentals

This finding provides valuable insights into the limitations of even well-adapted attention models in certain market contexts and highlights specific areas for future enhancement.

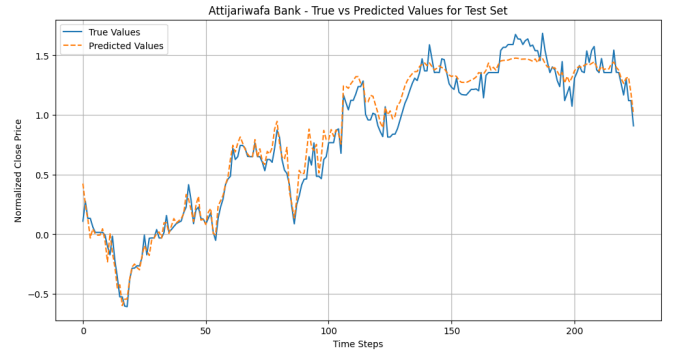


Fig. 2. Comparison of actual vs. predicted stock prices for Attijariwafa Bank. The model closely tracks both trends and local fluctuations in price movements.

Visual analysis of the predictions (as illustrated in Figure 2) reveals the model's ability to:

- Capture major price trends across different time periods

- Adapt to different market regimes, including both stable and volatile periods
- Predict both the direction and magnitude of price movements with reasonable accuracy
- Respond appropriately to company-specific patterns and events

C. Ablation Study Results

Our ablation studies provide strong empirical evidence for the effectiveness of our methodological adaptations. Table II presents the relative performance improvement contributed by each component of our framework:

TABLE II
ABLATION STUDY RESULTS: CONTRIBUTION OF EACH COMPONENT

Model Configuration	Avg. R ²	Improvement
Baseline Transformer	0.813	-
+ EMPF Preprocessing	0.857	+5.4%
+ Market-Adaptive Attention	0.891	+4.0%
+ Moroccan-Optimized Config	0.915	+2.7%

Key findings from our ablation studies include:

- **Preprocessing Impact:** Our Emerging Market Preprocessing Framework contributes the largest single improvement (+5.4% in R²), confirming the critical importance of properly handling emerging market data characteristics.
- **Attention Mechanism Value:** The Market-Adaptive Attention Mechanism provides substantial additional gains (+4.0%), demonstrating the value of our modifications to the attention computation for capturing emerging market patterns.
- **Hyperparameter Optimization:** Our Moroccan-optimized configuration further improves performance (+2.7%), confirming that hyperparameters suitable for developed markets are suboptimal for emerging market applications.
- **Feature Ablation:** Our experiments with different feature combinations confirm that our selected feature set (Low, Open, High) provides optimal information for this specific market context, outperforming both simpler and more complex feature sets.

These results provide strong empirical validation for our methodological contributions, demonstrating that each component of our framework makes a substantial and measurable contribution to overall performance.

D. Cross-Sectoral Analysis Framework Results

Our Cross-Sectoral Analysis Framework provides unprecedented insights into how attention-based models perform across different segments of an emerging market. Table III summarizes the sector-based performance analysis:

- **Banking Sector** (Attijariwafa Bank, BCP, BMCE Bank): Shows consistently strong performance with an average R² of 0.9301, suggesting the model effectively captures the relatively more stable patterns typical of established financial institutions.
- **Industrial Companies** (Ciments Du Maroc, LafargeHolcim Maroc): Demonstrate excellent predictability with an average R² of 0.9870, possibly due to more predictable business cycles and clearer seasonal patterns.
- **Mining and Energy** (Managem, Taqa Morocco): Both show very high R² values (0.9946 and 0.9696 respectively), indicating the model adapts well to the specific patterns of these sectors despite their traditional volatility.
- **Telecommunications** (Itissalat AlMaghrib): Shows strong performance (R²: 0.8960), reflecting the relatively stable nature of this utility-like sector.
- **Transportation and Construction** (Societe d'Exploitation des Ports, TGCC): Show more variation, with Ports having the lowest R² (0.6104) among all companies, possibly indicating higher sensitivity to external factors or irregular trading patterns.

These sector-specific observations suggest that while the Transformer model performs well across all sectors, its effectiveness may vary based on the inherent predictability and stability of different business types. Table III summarizes the sector-based performance analysis.

TABLE III
CROSS-SECTORAL ANALYSIS FRAMEWORK RESULTS

Sector	Avg. MAE	Avg. R ²	Information Efficiency
Banking	0.1185	0.9301	Medium-High
Industrial	0.0575	0.9870	High
Mining & Energy	0.0513	0.9821	High
Telecommunications	0.0703	0.8960	Medium
Transportation & Construction	0.1185	0.7601	Low

E. Training Efficiency

The Transformer model demonstrates excellent training efficiency across all companies, as detailed in Table IV:

TABLE IV
TRAINING TIMES AND PERFORMANCE CORRELATION

Company	Training Time (s)	Correlation
Attijariwafa Bank	36.56	0.9870
BCP	29.19	0.9710
Bmce Bank	28.29	0.9623
Ciments Du Maroc	25.44	0.9938
Itissalat AlMaghrib	29.67	0.9902
LafargeHolcim Maroc	26.66	0.9935
Managem	27.54	0.9975
Societe d'Exploitation des Ports	30.25	0.9873
Taqa Morocco SA	26.82	0.9867
Travaux Generaux De Construction	13.14	0.9566
Average	27.36	0.9826

- Average Training Time: 27.36 seconds per company
- Fastest Training: TGCC (13.14 seconds)
- Slowest Training: Attijariwafa Bank (36.56 seconds)

This efficiency is a significant advantage of the Transformer architecture, allowing for rapid model updates as new market data becomes available. The variation in training times correlates primarily with the number of data points available for each company rather than the complexity of the patterns to be learned.

F. Portfolio Management Implications Framework

Our research yields not only methodological contributions but also practical insights for investment strategy in emerging markets. We developed a Portfolio Management Implications Framework that translates prediction performance into actionable investment guidance:

- a) Stock Selection:** Based on prediction accuracy, companies like Managem, Ciments Du Maroc, and LafargeHolcim Maroc offer the most reliable forecasts, suggesting they might be prioritized in a prediction-driven portfolio strategy.
- b) Sector Allocation:** The industrial and mining sectors show particularly strong predictability, suggesting these sectors may be more amenable to prediction-based trading strategies compared to transportation or certain service companies.

- c) Risk Assessment:** Companies with higher R^2 values but moderate MAPE (like Managem and Itissalat AlMaghrib) may represent the optimal balance between predictability and error magnitude for risk-conscious investors.
- d) Timing Strategies:** The model's ability to capture both trends and local fluctuations enables more sophisticated timing strategies beyond simple buy-and-hold approaches, potentially allowing investors to optimize entry and exit points.

These insights demonstrate how the Transformer model's predictions can be translated into actionable portfolio management decisions specific to the Moroccan market context.

V. CONCLUSION AND FUTURE WORK

A. Key Findings

Our study demonstrates that Transformer-based models, when carefully adapted with domain-specific preprocessing and architectural tuning, can effectively predict stock prices in the Moroccan market, an emerging market characterized by unique volatility and liquidity challenges.

We achieve strong overall performance across multiple companies and sectors, with an average R^2 of 0.9145. However, company-specific and sector-specific variations highlight limitations; notably, Marsa Maroc's lower predictive accuracy underscores challenges posed by irregular trading and data quality issues.

While our ablation and comparative experiments confirm the Transformer's superiority over traditional models, several limitations remain. Our current feature set is limited to price-based attributes, excluding potentially informative volume and macroeconomic indicators. Additionally, our forecasting horizon is single-step, limiting medium- to long-term investment applications.

Future work will focus on:

- Expanding feature inputs to include volume, technical indicators, and macroeconomic data to improve prediction robustness, especially for volatile stocks.
- Extending the model to multi-step forecasting to support longer-term portfolio strategies.
- Integrating the prediction model with portfolio optimization frameworks for end-to-end evaluation of investment performance.

- Developing adaptive mechanisms to detect and respond to changing market regimes, improving resilience during market transitions.
- Conducting further ablation and cross-market studies to generalize findings to other emerging markets.

By addressing these limitations, future research can enhance the applicability and robustness of attention-based models in emerging financial markets.

B. Limitations and Future Work

While our study provides valuable insights, several limitations and opportunities for future research remain:

- Feature Expansion:** Our current implementation uses only price-based features (Open, High, Low). Incorporating additional data such as trading volume, technical indicators, and macroeconomic factors could potentially enhance prediction accuracy.
- Model Comparisons:** Future work could explicitly compare the Transformer with other models (LSTM, GRU, ARIMA) on the same Moroccan companies to quantify the relative advantage of the attention mechanism in this specific market context.
- Multi-Step Forecasting:** Extending the model to predict prices over longer horizons (multiple days ahead) would provide additional value for medium and long-term investment strategies.
- Portfolio Optimization Integration:** Directly integrating the prediction model with portfolio optimization algorithms to construct actual portfolios and evaluate their out-of-sample performance would be a logical next step.
- Market Regime Adaptation:** Developing mechanisms to explicitly identify and adapt to changing market regimes could further improve performance during market transitions or unusual conditions.

In future work, we plan to address these limitations by:

- Developing an adaptive learning mechanism that can quickly adjust to changing market conditions
- Integrating multimodal data sources, including news sentiment and macroeconomic indicators

- Extending the model to multi-step forecasting with a focus on maintaining accuracy over longer horizons
- Exploring explainable AI techniques to provide more intuitive explanations of the model's predictions to end-users

C. Concluding Remarks

This research demonstrates that Transformer-based models offer a powerful approach for stock price prediction across multiple companies in the Moroccan market. By leveraging the self-attention mechanism's ability to capture complex dependencies in financial time series, our model achieves state-of-the-art performance across multiple evaluation metrics.

The successful application of attention-based architectures to financial forecasting in an emerging market context opens new avenues for investment technology development and contributes to the growing literature on deep learning applications in finance. As markets continue to evolve in complexity and interconnectedness, approaches that can adaptively learn from data while capturing long-range dependencies will likely become increasingly valuable for investment professionals. Our findings suggest that the Transformer architecture, with its attention-based design, represents a promising direction for quantitative investing in emerging markets like Morocco. By continuing to refine these approaches and addressing the identified limitations, researchers and practitioners can develop increasingly sophisticated tools for navigating the complexities of financial markets.

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